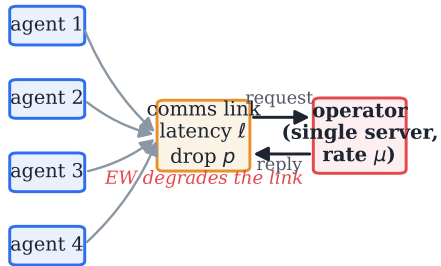
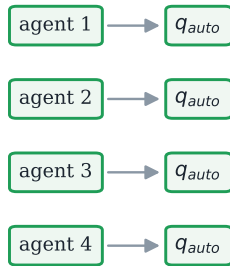


Direct control (human in the loop)



resulting control quality
served in time: q_{op} · waiting: q_{stall}
dropped / timeout: $q_{fallback}$

Supervisory autonomy (on the loop)



decides locally, every step
no queue, no link: steady q_{auto}
comms-invariant

attention dilution: $q_{op}^{\text{eff}} = q_{auto} + (q_{operator} - q_{auto}) \min(1, \kappa/n)$. More agents n dilute the per-decision quality and lengthen the queue.